Bose–Hubbard NQS
Periodic 2D systems

ED construction • supervised representation • small-system VMC

10-31-2026

What are we testing?

1. Is the Bose–Hubbard ED construction correct?

2. Can basic NQS reproduce its ground state with supervision?

3. Does stochastic VMC work on small, checkable systems?

4. What does a fixed-N fit predict in untrained sectors?

Building and benchmarking a physics tool. Architecture searches and optimizer rankings are separate research directions.

1. Exact-diagonalization reference

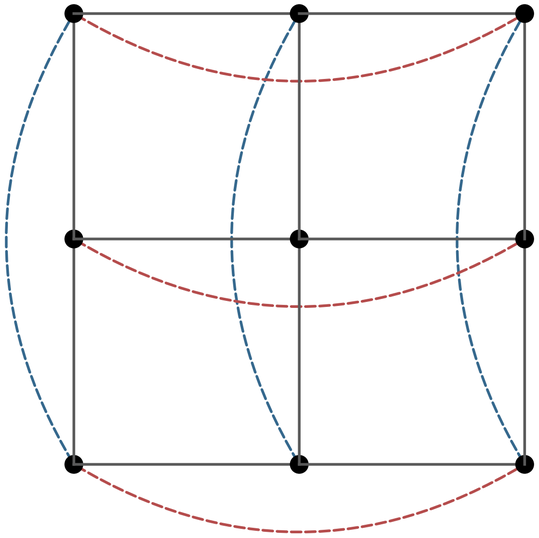
Hamiltonian

$$H = -t \sum_{\langle i,j \rangle} (b_i^\dagger b_j + b_j^\dagger b_i) + \frac{U}{2} \sum_i n_i(n_i - 1) - \mu \sum_i n_i$$

$$t = 1, \quad U = 16.74, \quad \mu = 0$$

$$N_{\text{tot}} = L_x L_y, \quad 0 \le n_i \le 4$$

Periodic in both x and y (torus)



y: periodic

x: periodic

Main-body systems are periodic in BOTH directions; the diagram is a3×3 torus. Cylinder results remain in supporting material. All hopping bonds are unique unordered pairs; short length-two axes have one bond per pair.

Independent ED and DMRG check

Quantity	3×3 torus
Basis dimension / unique bonds	19,855 / 18
Independent ED ground energy / t	-6.104592578442
Independent ED residual norm	8.3e-14 t
DMRG energy error: max bond128	1.93e-07 t
DMRG energy error: max bond256	4.59e-10 t
Max full one-body error: max bond256	3.54e-10
Max site-density error: max bond256	1.21e-11

Both axes periodic; N=9, cutoff4, t=1, U=16.74. DMRG uses a periodic Hamiltonian with an open-chain MPS representation. Requested maximum bond dimensions128 and256 are checked against ED; the physical boundaries are never opened.

Density and nearest-neighbor coherence

$$\rho_i = \langle n_i \rangle, \qquad G_{ij} = \langle b_i^\dagger b_j \rangle, \qquad g_{ij} = \mathrm{Re}\, G_{ij}$$

C is a chosen set of physical lattice sites; α denotes x or y.

$$B_\alpha(C) = \{\{i,j\} \in E_\alpha : \; i \in C, \; j \in C\}$$

Eα contains unique unordered nearest-neighbor bonds along α, including wraps.

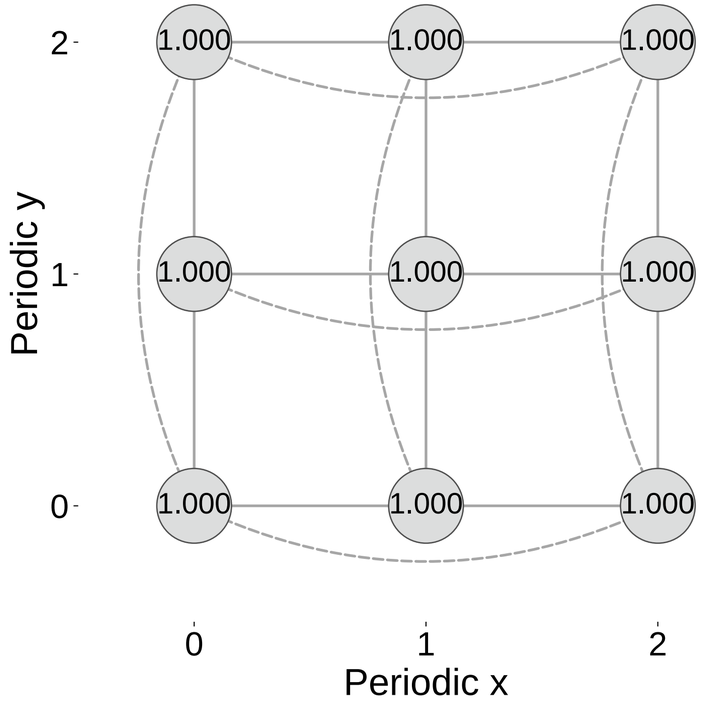
$$\overline{G}_\alpha(C) = \frac{1}{|B_\alpha(C)|} \sum_{\{i,j\} \in B_\alpha(C)} g_{ij}$$

Main3×3 torus: C = all nine sites; |Bx(C)| = |By(C)| = 9.

A bond belongs to Bα(C) only when BOTH endpoints lie in C. Each physical edge is counted once. Coherence is a one-body correlation, not <n_i n_j>. At fixed N, <b_i>=0 and sum_i rho_i=N; uniform density alone does not establish superfluidity.

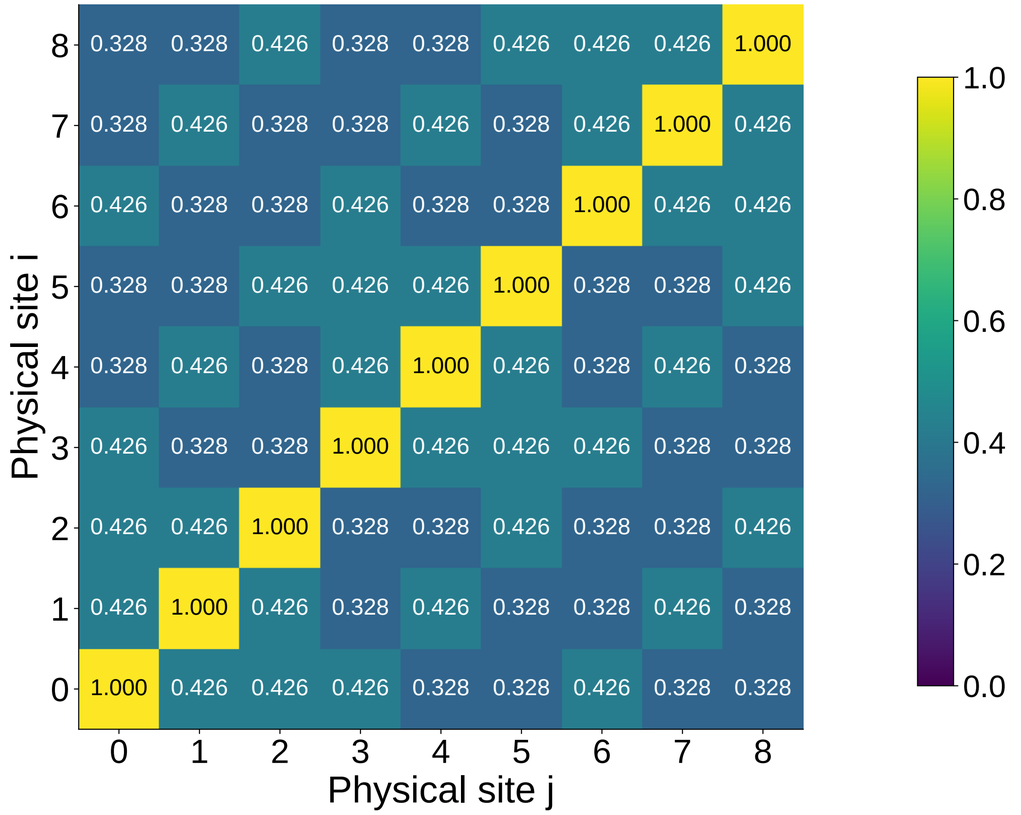
ED ground state: density and coherence

Site density $\rho_i = \langle n_i \rangle$



Uniform mean density; not one occupation configuration.

One-body matrix $G_{ij} = \langle b_i^\dagger b_j \rangle$



NN means: Gx = 0.425670, Gy = 0.425670

Actual3×3-torus ED ground state: N=9, cutoff4, t=1, U=16.74. Full19,855-state sums; site i=yLx+x. Translation symmetry gives rho_i=1; numerical roundoff is not a density texture. The right panel is the complete one-body matrix, with diagonal Gii=rho_i. Nearest-neighbor averages include all nine bonds per direction; nonzero coherence alone does not establish thermodynamic order.

2. Supervised NQS

Five basic ansatzes

FC	Conv2D	Jastrow	RBM	Transformer
One-hot input	One-hot grid	Onsite tables	One-hot input	Occupation + (x,y)
↓	↓	↓	↓	↓
Tanh dense layers	3×3 Conv + SiLU	Shared pair tables	Softplus hidden sum	One attention block
↓	↓	↓	↓	↓
Scalar log amplitude	Dense scalar head	Sum log amplitude	Bias + scalar	Mean pool + scalar
No imposed lattice symmetry.	Boundary-aware padding. Global dense head.	Fixed physical baseline.	Binary hidden units.	ReLU, one head; no dropout.

All five ansatzes represent real positive amplitudes for the untwisted hopping model.

Supervised objective

$$\tilde{f}_\theta(\mathbf{n}) = f_\theta(\mathbf{n}) - \frac{1}{2}\log \sum_{\mathbf{n}\in\mathcal{H}_N} e^{2f_\theta(\mathbf{n})}$$

$$\mathcal{L}_{\log} = \frac{1}{D_N} \sum_{\mathbf{n}\in\mathcal{H}_N} [\tilde{f}_\theta(\mathbf{n}) - \log \Psi_{\mathrm{ED}}(\mathbf{n})]^2$$

$$F_{\mathrm{abs}} = |\langle \Psi_\theta | \Psi_{\mathrm{ED}} \rangle|, \qquad \mathcal{L}_{\mathrm{infidelity}} = 1 - F_{\mathrm{abs}}^2$$

Log-MSE is primary; infidelity refinement is a separate diagnostic.

Main fits use Adam + log-MSE over the chosen particle sector only. Reported overlaps and energies are checked against the raw ED ground vector. Unseen particle-number sectors never enter this loss.

Can basic NQS fit the ED state?

Ansatz	2×3 Fabs	3×3 Fabs	3×3 δE/site
FC	0.999998	0.997320	0.01174
Conv2D	0.999998	0.997997	0.008611
RBM	0.999947	0.995000	0.02035
Transformer	0.999995	0.997980	0.009526
Jastrow	0.997298	0.992349	0.02869

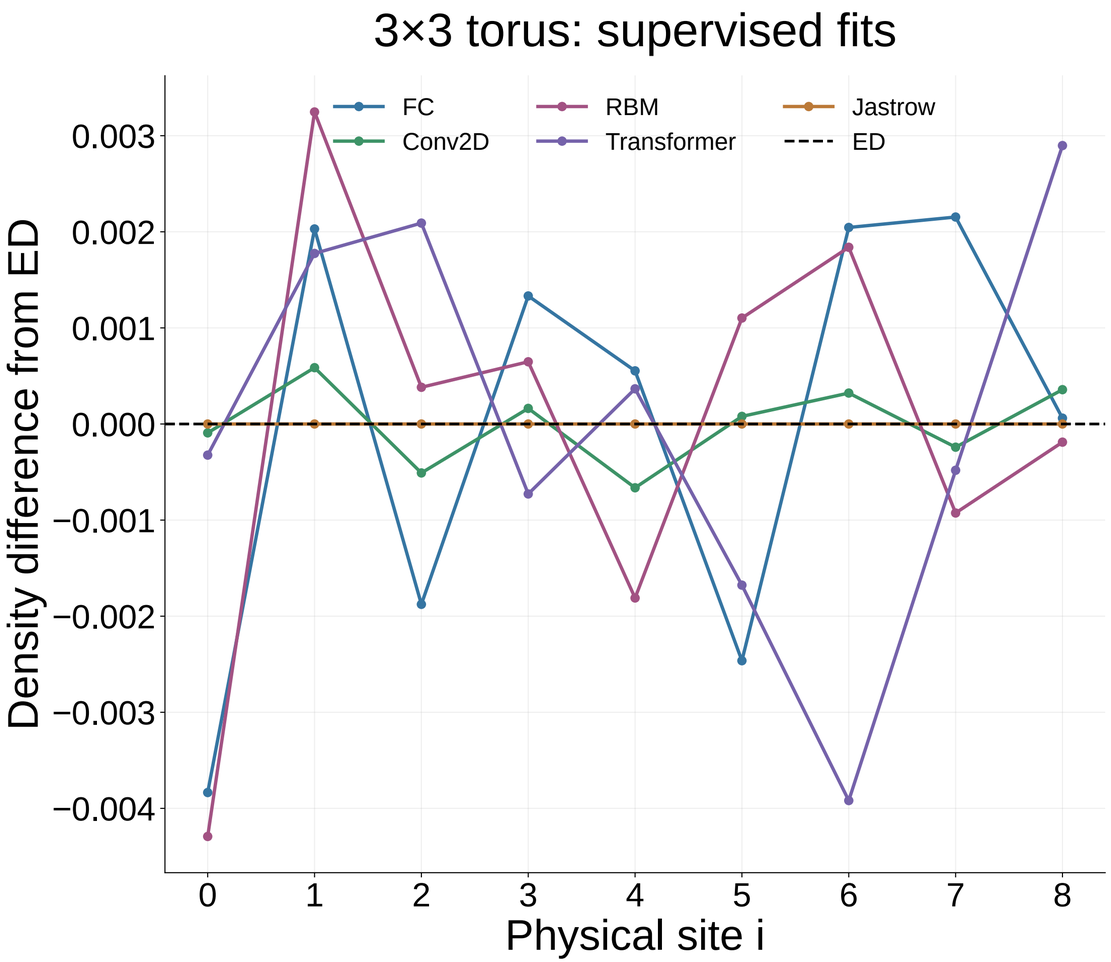
ALL results are periodic in both axes. Adam + normalized log-MSE; checkpoint selected by log-MSE. Four neural fits reach Fabs≥0.9999 on2×3 and≥0.99 on3×3. Width32, actual trained primary fits. RBM tiny fit20000updates, others10000; Jastrow is a physical baseline.

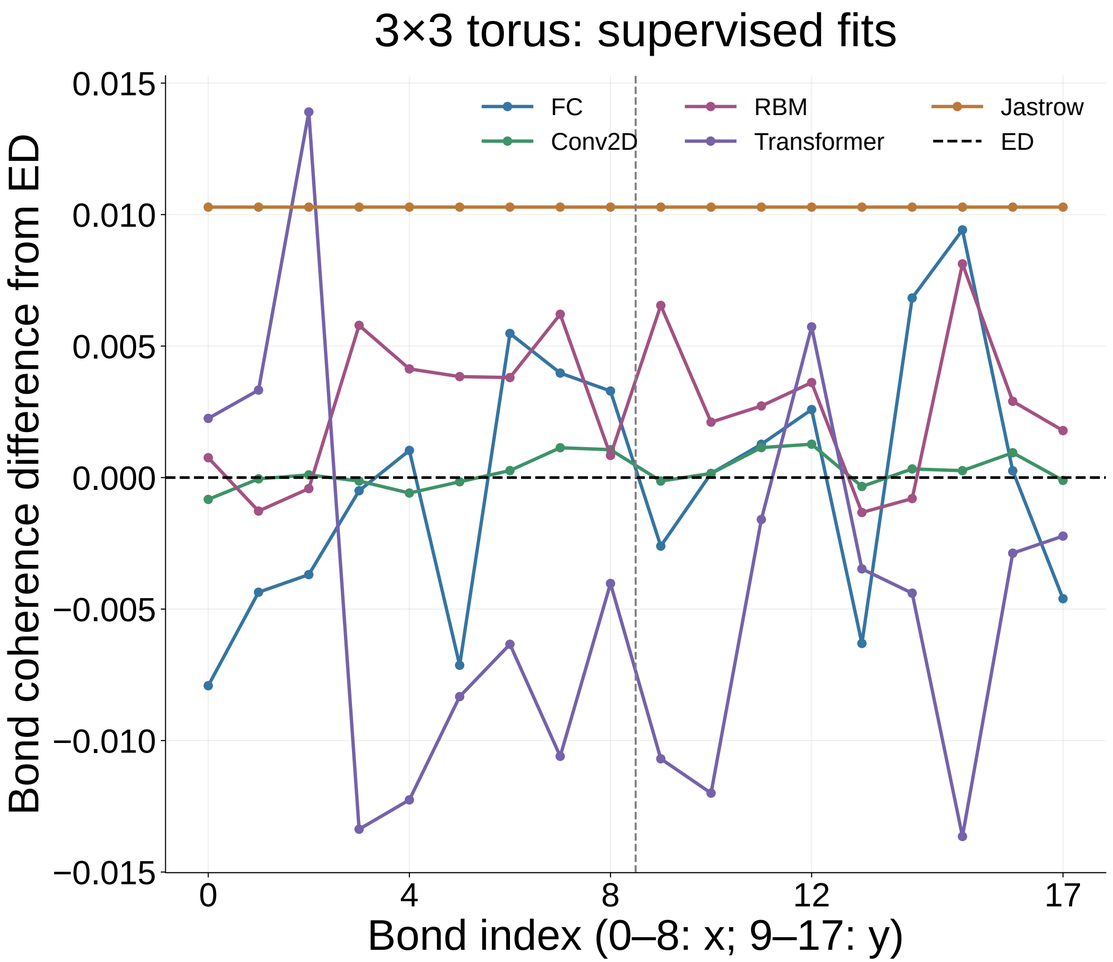
Supervised fits: local observables

3×3 torus	max $\|\delta\rho_i\|$	Mean D	Gx	Gy
ED	0	0.122388	0.425670	0.425670
FC	0.00384	0.123716	0.424578	0.426446
Conv2D	0.000664	0.123532	0.425759	0.426061
RBM	0.00429	0.126130	0.428300	0.428522
Transformer	0.00392	0.121386	0.421732	0.420651
Jastrow	7.11e-15	0.130732	0.435954	0.435954

Exact CANONICAL ED and NQS expectations, not MC. Density is compared SITE BY SITE; its global mean is fixed to1. D=<n(n−1)> is an onsite factorial moment. Gx/Gy average all nine unique bonds per direction, including periodic wraps.

Supervised states: density and coherence





3×3 torus, N=9, cutoff4. Full fixed-sector enumeration for ED and each supervised NQS: no Monte Carlo. Left: all nine site densities minus ED. Right: all18 unique NN bond coherences minus ED, ordered x then y with wraps included. These residuals measure state approximation error.

3. Small-system VMC

Variational Monte Carlo

$$E_\theta = \sum_{\mathbf{n} \in \mathcal{H}_N} P_{\theta,N}(\mathbf{n}) E_{\mathrm{loc}}(\mathbf{n})$$

$$E_{\mathrm{loc}}(\mathbf{n}) = \sum_{\mathbf{n}'} H_{\mathbf{n},\mathbf{n}'} \frac{\Psi_\theta(\mathbf{n}')}{\Psi_\theta(\mathbf{n})}$$

$$\nabla_\theta E = 2\left\langle (E_{\mathrm{loc}} - \overline{E}) \nabla_\theta \log \Psi_\theta \right\rangle$$

Equilibrate fresh walkers before updates; assess frozen checkpoints.

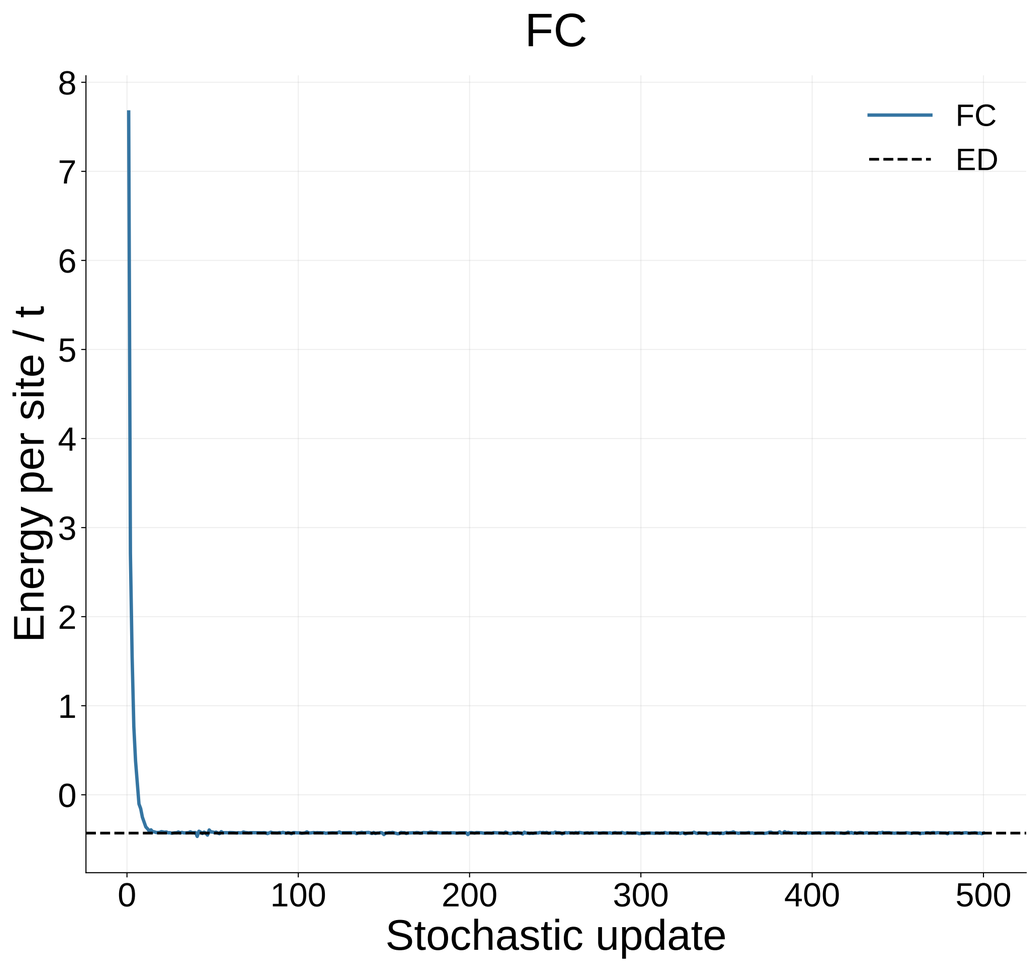
Retained configurations are correlated. Proposal spacing, burn-in, independent evaluation chains and time-block statistics are recorded.

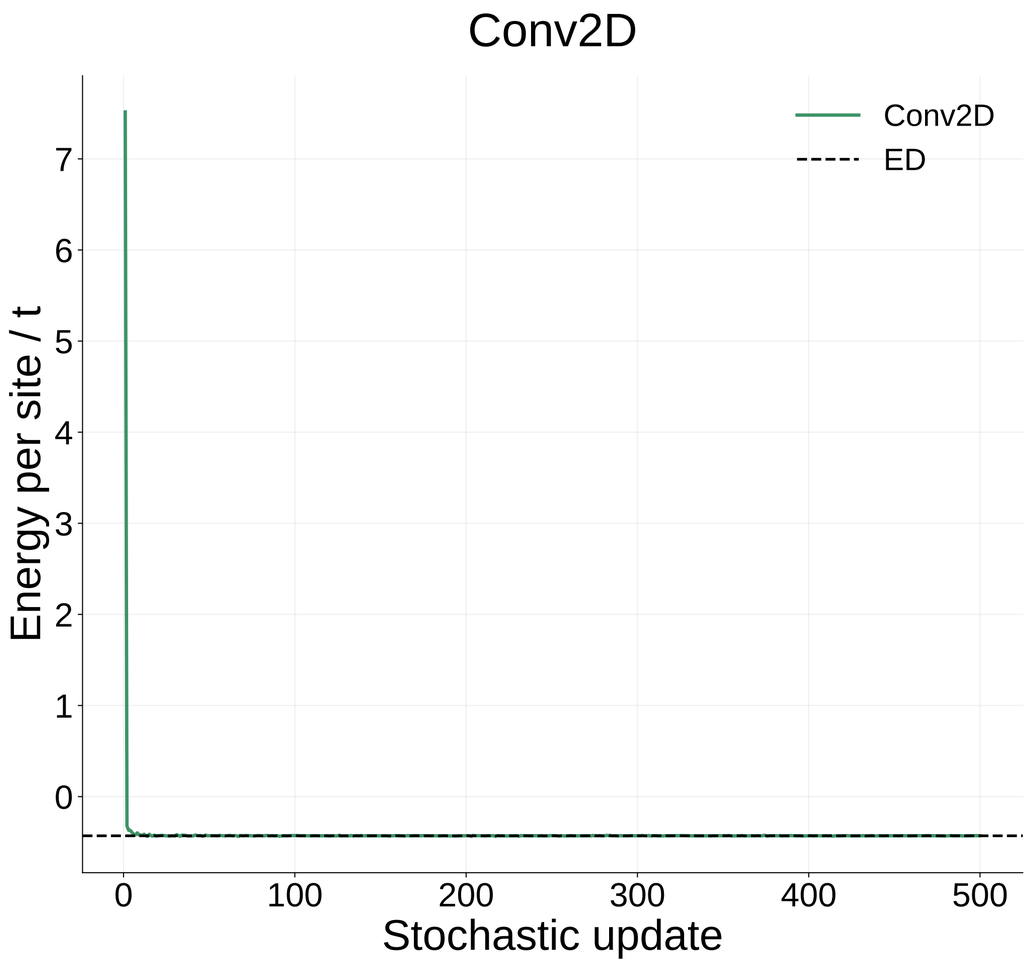
Small-system VMC versus ED

Ansatz	Initial E/t	Final E/t	Final Fabs	Var(H)/t²
FC	45.2128	-2.573497	0.999919	0.3786
Conv2D	45.5074	-2.579806	0.999984	0.0603

2×3 torus, N=6, cutoff4, 426 states. ED E/t=-2.581089512. Fresh initialization;500 stochastic SR+nativeSGD updates, no ED labels or supervised warm start. Exact final-state metrics; variance remains nonzero.

VMC trajectory and independent checks





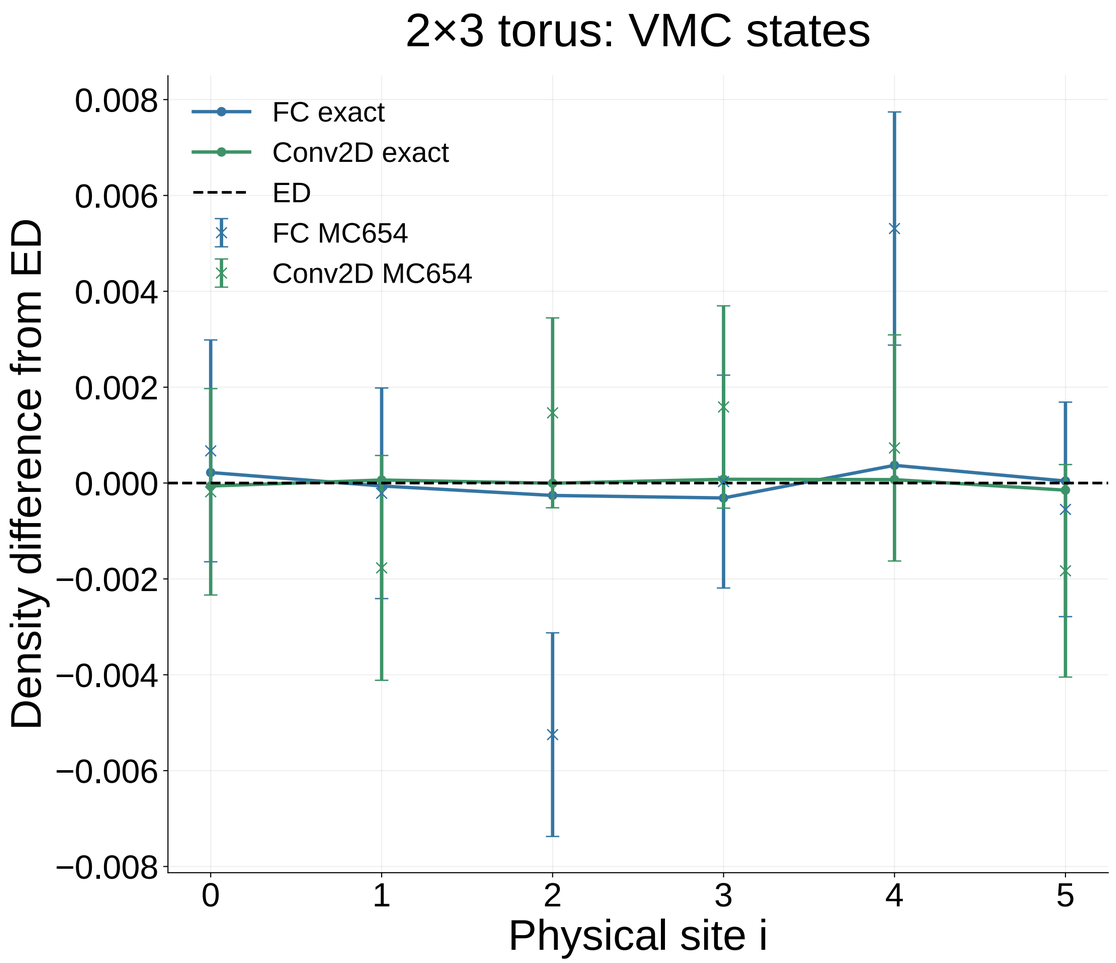
Curves are noisy training Monte Carlo estimates before each update, not exact energies or monotone descent. Exact final metrics and frozen321/654 measurements are assessed separately.
One fit per model is a proof of principle, not an architecture ranking.

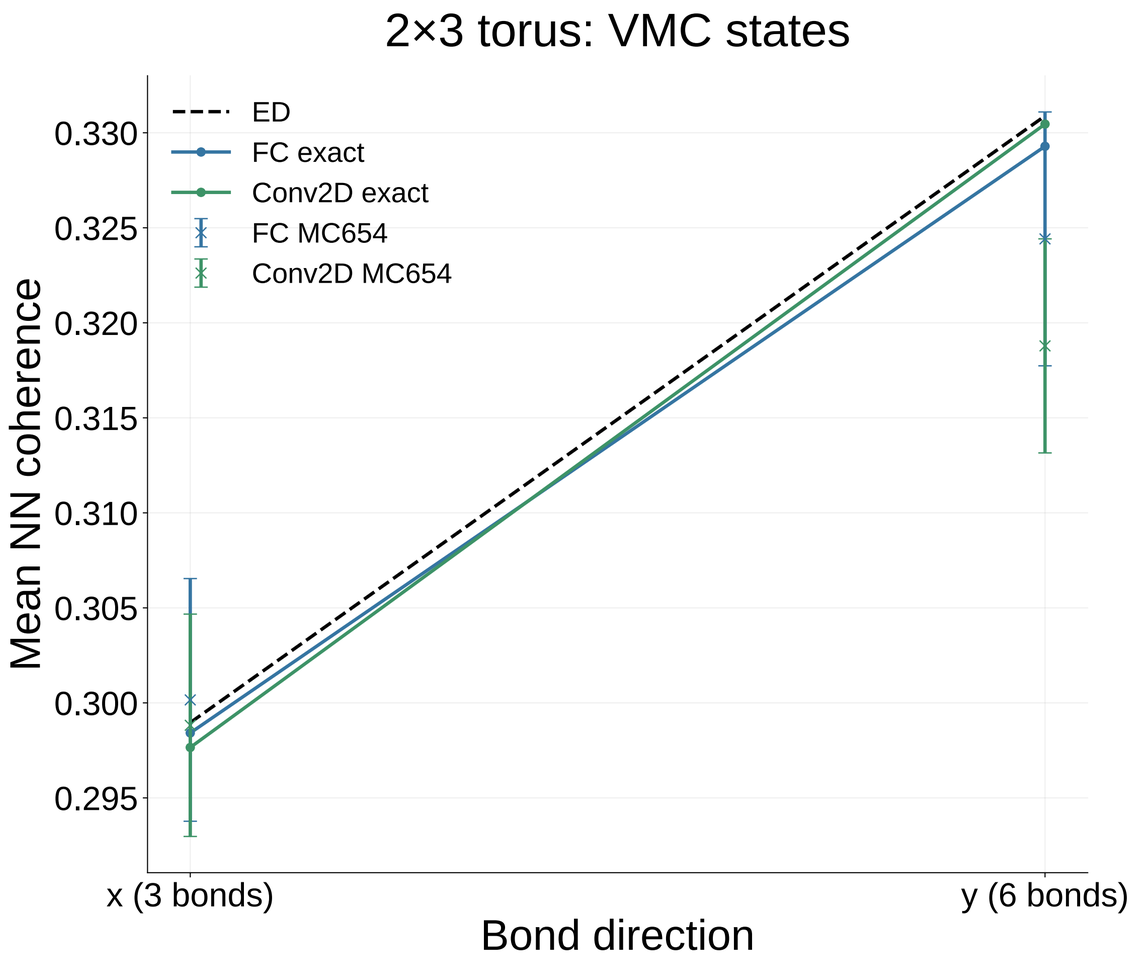
Frozen sampler versus exact state

Ansatz / seed	MC E/t ± conditional SEM	Exact saved-state E/t
FC / 321	-2.57750 ± 0.00573	-2.573497
FC / 654	-2.57818 ± 0.00391	-2.573497
Conv2D / 321	-2.57907 ± 0.00152	-2.579806
Conv2D / 654	-2.58025 ± 0.00135	-2.579806

Both evaluation seeds are retained; neither selects the training checkpoint. Error estimates are conditional on mixing and stationarity, not confidence intervals or proof of equilibration.

VMC states: density and coherence





2×3 torus, N=6, cutoff4. Exact saved-NQS expectations use all426 configurations; frozenMC654 crosses show independent sampled estimates ± conditionalSEM. ED is also enumerated exactly. Both seeds appear in the appendix. Three x and six y bonds: this short rectangle need not have Gx=Gy.

Criticality and finite-size validation

$$\Delta_N = E_1(N) - E_0(N), \qquad F_{\mathrm{abs}} = |\langle \Psi_0 | \Psi_\theta \rangle|$$

$$E_\theta - E_0 \geq \Delta_N (1 - F_{\mathrm{abs}}^2)$$

A closing gap weakens energy as a certificate of fidelity.

Gaplessness does not imply that variational Monte Carlo must fail.

Finite tori require their own checks before bulk claims.

Bound assumes a unique ground state and a normalized trial state in the same sector. Use total expectation energy, not energy density or one noisy MC estimate. Carleo & Troyer, Science355,602(2017), demonstrate finite-size NQS-VMC at criticality; no thermodynamic guarantee follows.

Neutral gap: excitation at fixed N

$\Delta_N = E_1(N) - E_0(N)$: lowest two eigenvalues of the SAME fixed-N Hamiltonian.

Geometry / sector	E0/t	Neutral gap ΔN/t
2×3 torus / N=6	-2.581090	11.075543
3×3 torus / N=9	-6.104593	9.015297

Neutral means no particle is added or removed: E1(N) is the first excited energy in the same sector as E0(N). This is an ED spectral gap, not a neural-network metric. The charge gap E0(N+1)+E0(N−1)−2E0(N) is different. Both eigenpairs are checked by residuals; these finite clusters do not establish an infinite-system gap.

4. Particle-sector leakage

Fixed-N fit and untrained particle sectors

$$\widetilde{\psi}_\theta(\mathbf{n}) = e^{f_\theta(\mathbf{n})}, \qquad Z_N = \sum_{\sum_i n_i = N} e^{2f_\theta(\mathbf{n})}$$

$$\psi_{\theta,N_0}(\mathbf{n}) = \delta_{\sum_i n_i, N_0}\, e^{f_\theta(\mathbf{n})}/\sqrt{Z_{N_0}}$$

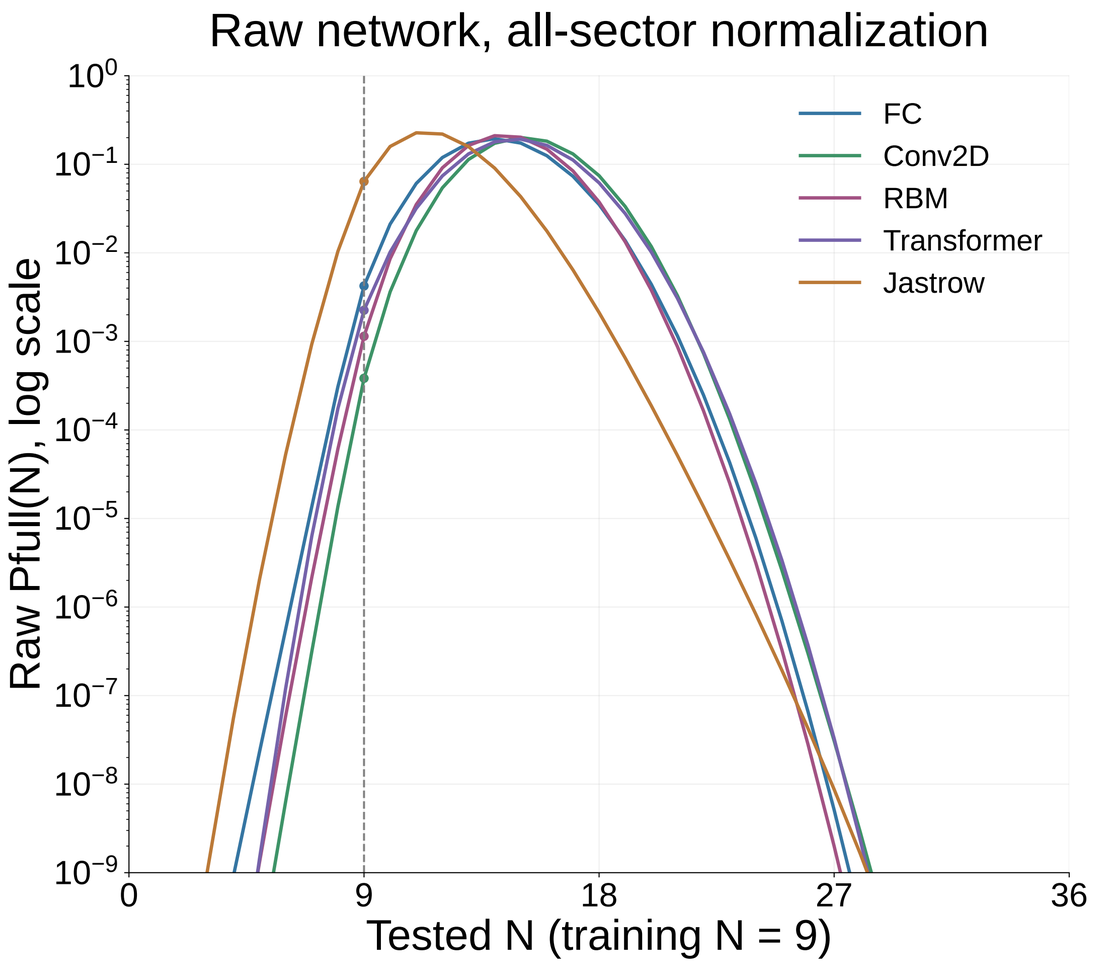
$$P_{\text{full}}(N) = Z_N / \sum_{K=0}^{36} Z_K, \qquad \ell_{\text{full}} = 1 - P_{\text{full}}(N_0)$$

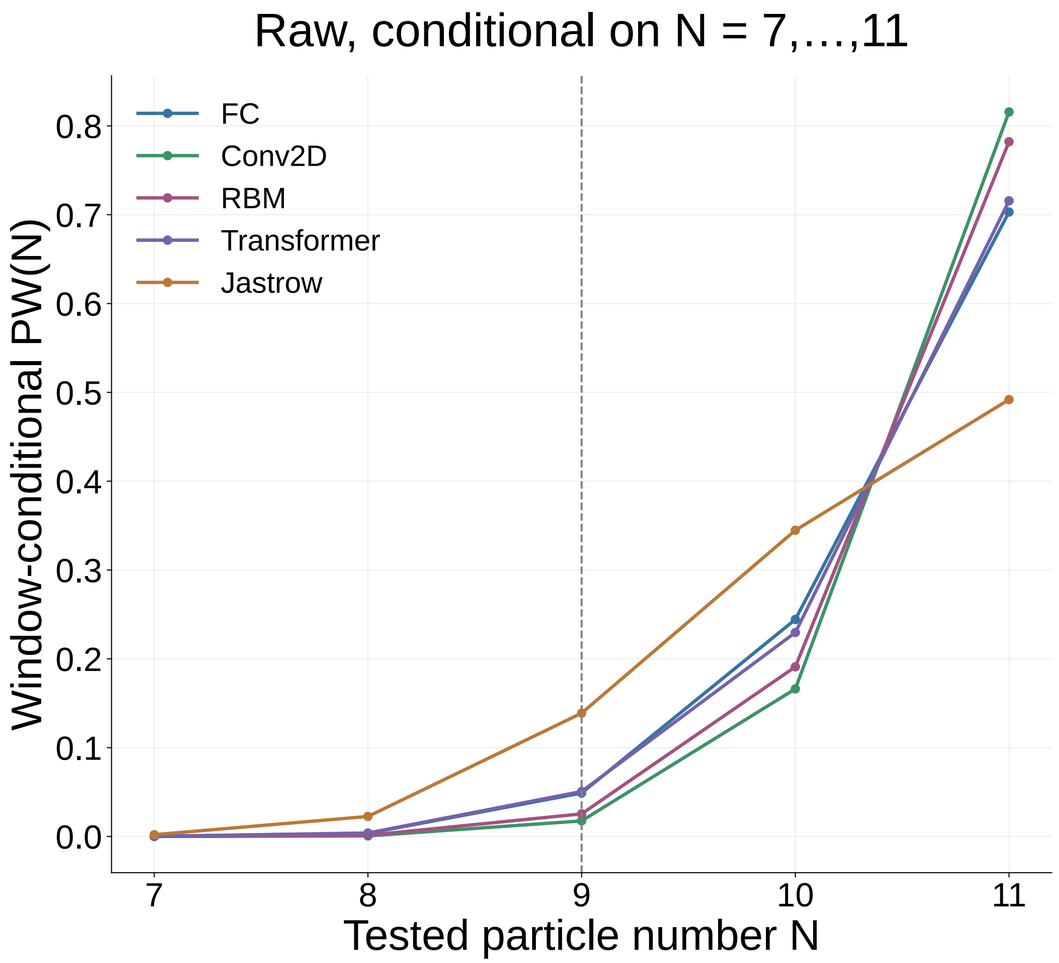
$$W = \{7, 8, 9, 10, 11\}, \qquad P_W(N) = Z_N / \sum_{K \in W} Z_K$$

Training sees only N0=9; the raw network also returns values outside it.

Nine sites with local occupations0,…,4 allow N=0,…,36. Full probabilities normalize over all5^9 configurations; PW conditions only on W. The physical canonical state is zero outside N0; the raw exp(f) generally is not. Raw leakage tests an untrained extension, not the particle numbers visited by canonical VMC.

Measured raw sector weights





Same supervised3×3-torus primary fits, trained ONLY at N=9. All5^9 configurations are enumerated. LEFT: raw globally normalized probabilities, with dots at N=9 and tails below10^-9 clipped. RIGHT: the same raw weights conditioned on N=7,…,11. High canonical overlap need not imply large raw mass at N=9; these are different normalizations.

Leakage and within-sector accuracy

Ansatz	F9	Pfull(9)	Nearby leakage	Total leakage
FC	0.997320	0.004227	8.2189%	99.577291%
Conv2D	0.997997	0.0003831	2.1456%	99.961694%
RBM	0.995000	0.001142	4.3750%	99.885770%
Transformer	0.997980	0.00225	4.2179%	99.774996%
Jastrow	0.992349	0.06401	39.7089%	93.598656%

3×3 TORUS, cutoff4. Nearby leakage sums GLOBAL probabilities of N=7,8,10,11; total leakage sums every N≠9. These are not window-conditional probabilities. High canonical overlap can coexist with large raw leakage; no universal architecture ranking follows.

Fixed-sector data cannot fix leakage

$$f'_\theta(\mathbf{n}) = f_\theta(\mathbf{n}) + h(N), \qquad h(N_0) = 0$$

$$Z'_N = e^{2h(N)} Z_N, \qquad \Psi'_{\theta, N_0} = \Psi_{\theta, N_0}$$

All canonical fits and observables stay unchanged.

RBM visible biases implement a number tilt within the same family.

Neighboring-sector weights depend on an untrained extension.

This is an exact postprocessing control, not a claim that each finite neural architecture realizes every h. Examples: h=a(N−N0) tilts sectors; h=−lambda(N−N0)^2 suppresses both sides. Distinct projected particle-number states are orthogonal; these weights are not transition probabilities.

Same canonical state, different leakage

Postprocessing of FC log amplitude	Pfull(9)	F9	Canonical energy
Raw output	0.0042271	0.997320	unchanged
$h = -0.5(N-9)$	0.13144	0.997320	unchanged
$h = +0.5(N-9)$	2.4929e-06	0.997320	unchanged
$h = -(N-9)^2$	0.59147	0.997320	unchanged

Algebraic reweighting of the SAME measured logZN, without new inference, fitting or checkpoint selection. h(9)=0 throughout. Changes in raw full-space weight cannot be detected by any fixed-N9 fitting metric.

Enforce the intended physical support

$$\Psi_\theta^{\text{projected}} \propto P_{N_0} \widetilde{\Psi}_\theta$$

$$P^{\text{projected}}(N) = \delta_{N,N_0}, \quad \ell_{\text{full}}^{\text{projected}} = 0, \quad F^{\text{projected}} = F_{N_0}$$

No retraining is needed; canonical predictions are preserved.

A number penalty on fixed-N training samples is identically zero.

Restricting sampled configurations is standard for U(1) conservation in canonical VMC (Pei & Clark2024). Learning an off-sector penalty requires off-sector examples or sampling; explicit output masks can instead impose it directly. Raw leakage is an extension diagnostic, not an error of projected VMC or proof of wasted capacity.

What this benchmark establishes

ED: independent physics and numerical checks agree.

Supervision: basic neural ansatzes approximate the ED state.

VMC: validate stochastic optimization on small finite clusters.

Raw neighboring-sector amplitudes are a separate diagnostic.

Exact number projection enforces the intended physical support.

Large-system critical performance and improved NQS optimization remain separate questions. Neither leakage nor a finite-budget VMC plateau by itself proves an expressiveness limit.

Supporting calculations

SR derivation, sampling checks and cylinder context

Appendix

Validate the ED construction

Independent check	What it establishes
Atomic limit: $t = 0$	$U\, n(n-1)/2$ diagonal and fixed-N chemical shift
One particle,3×3 torus	$E0/t = -4$ (uniform degree-four hopping graph)
Two particles / two sites	$E0 = (U-\sqrt{(U^2+16t^2)})/2$ (one hopping bond)
Explicit ladder operators	Independent 2×2 Hamiltonian and bosonic factors
Dense versus sparse ED	Small-system spectrum and eigenvector residual
Number and boundaries	Canonical basis, wraps, no self/doubled bonds

15 ED-specific tests pass; the periodic single-particle result is checked separately. Independent operator construction tests the physics conventions; solver residual alone would not validate the Hamiltonian.

Exact measurement on small systems

$$p_N(\mathbf{n}) = |\psi_N(\mathbf{n})|^2, \qquad \sum_{\mathbf{n} \in \mathcal{H}_N} p_N(\mathbf{n}) = 1$$

$$\rho_i = \sum_{\mathbf{n} \in \mathcal{H}_N} p_N(\mathbf{n})\, n_i$$

$$G_{ij} = \sum_{\mathbf{n}, \mathbf{m} \in \mathcal{H}_N} \psi_N^*(\mathbf{n}) \langle \mathbf{n} | b_i^\dagger b_j | \mathbf{m} \rangle \psi_N(\mathbf{m})$$

Enumerate all fixed-N configurations for BOTH ED and the saved NQS.

2×3: 426 states; 3×3: 19,855 states (unit filling, cutoff4).

These are full-sector expectations without sampling error. Off-diagonal matrix elements connect occupations by one boson hop and include the bosonic square-root factor. Exact saved-NQS error versus ED measures the state approximation; it does not test the sampler.

Monte Carlo: density and coherence

$$\mathbf{n}^{(m)} \sim p_{\theta,N_0}, \qquad \langle O \rangle \simeq M^{-1} \sum_m O_{\mathrm{loc}}(\mathbf{n}^{(m)})$$

$$O_{\mathrm{loc}}(\mathbf{n}) = \sum_{\mathbf{m}} \langle \mathbf{n}|O|\mathbf{m}\rangle \, \psi_\theta(\mathbf{m})/\psi_\theta(\mathbf{n})$$

$$\rho_i \simeq M^{-1} \sum_m n_i^{(m)}, \qquad D_i \simeq M^{-1} \sum_m n_i^{(m)}(n_i^{(m)} - 1)$$

$$R_{ij}(\mathbf{n}) = \sqrt{n_i(n_j+1)}\, e^{f_\theta(\mathbf{n}-\mathbf{e}_i+\mathbf{e}_j)-f_\theta(\mathbf{n})}, \quad i \neq j$$

$$G_{ij} \simeq M^{-1} \sum_m R_{ij}(\mathbf{n}^{(m)}), \qquad G_{ii} = \rho_i$$

Row-action convention: the sampled configuration is the bra; the connected ket removes one boson at i and adds it at j. Set R=0 unless n_i>0 and n_j<Nmax. Wavefunctions here are real and positive; each hop preserves total N. D_i is a factorial moment; Var(n_i)=D_i+rho_i−rho_i².

Frozen measurements and error bars

Freeze the checkpoint; sample fixed-N chains with no parameter updates.

$$g_{ij}^{(m)} = \tfrac{1}{2}(R_{ij}^{(m)} + R_{ji}^{(m)}), \qquad \overline{G}_{\alpha}^{(m)} = |B_{\alpha}(C)|^{-1} \sum_{\{i,j\} \in B_{\alpha}(C)} g_{ij}^{(m)}$$

Form each site/bond average PER SAMPLE, retaining its covariance.

Block consecutive times within each walker before pooling block means.

$$\mathrm{SEM}_{\mathrm{shown}} = \max(\mathrm{SEM}_{\mathrm{blocks}}, \mathrm{SEM}_{\mathrm{walker\ means}})$$

Frozen seeds321/654 are both retained; plots use predeclared654. Small2×3 cases:64walkers ×512retained steps, burn4096 proposals/walker, thin8, block64; new6×4 rows:128walkers, thin16. A proposal is one local move attempt, not a lattice sweep. SEM assumes stationarity, independent walkers and sufficiently decorrelated blocks; it is not proof of mixing. Larger NQS cylinders use MC versus DMRG, not enumeration.

Sector normalizations and overlap

$$\ell_{\text{near}} = \sum_{N \in W,\ N \neq N_0} P_{\text{full}}(N), \qquad \ell_{\text{full}} = 1 - P_{\text{full}}(N_0)$$

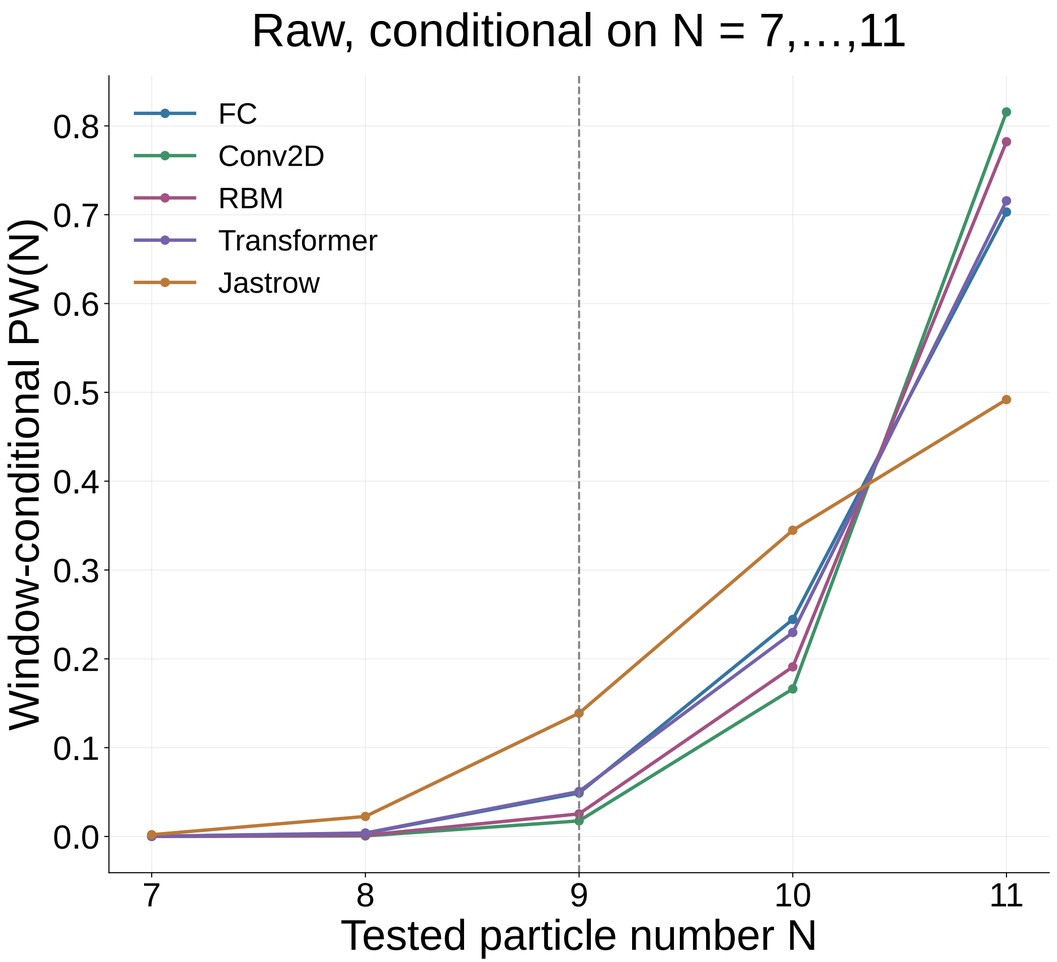
$$d_N = \dim \mathcal{H}_N, \qquad \overline{w}_N = Z_N / d_N, \qquad R_N = Z_N / Z_{N_0}$$

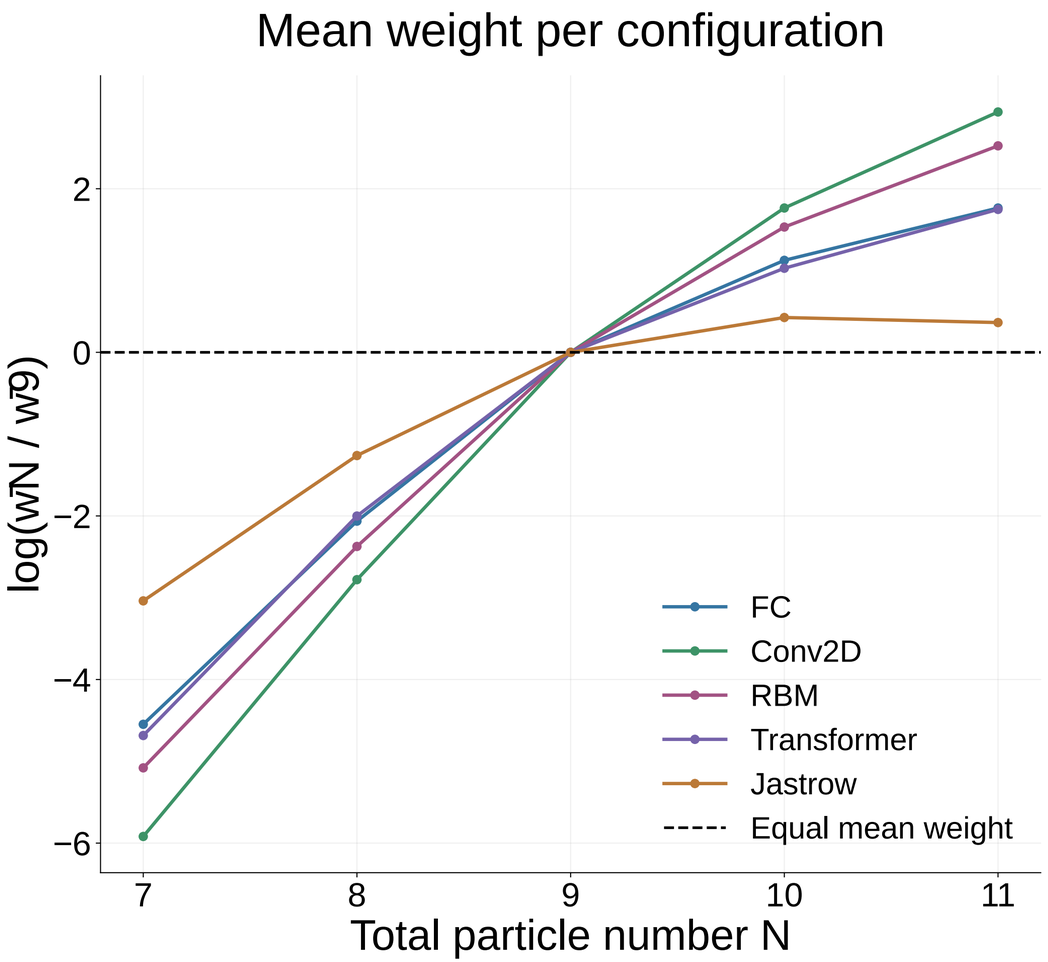
$$F_{\text{full}} = \sqrt{P_{\text{full}}(N_0)}\, F_{N_0}$$

Ffull compares the raw full-space state to zero-embedded canonical ED.

All overlaps are absolute and unsquared. Full means the finite cutoff0..4 basis; it does not mean unbounded Fock space. PW is conditional on the nearby window and cannot substitute for Pfull in either leakage definition.

Nearby untrained sectors: two diagnostics





Both panels use UNPROJECTED3×3-torus outputs. LEFT renormalizes over N=7,…,11 only, not the full space or the physical fixed-N state. RIGHT removes sector multiplicity using w̄N=ZN/dN; d7=6,030 and d11=48,879. Dividing by w̄9 removes the overall amplitude scale.

Ridge SR: definitions

$$\Psi_\theta(\mathbf{n}) = e^{f_\theta(\mathbf{n})} / \sqrt{\sum_{\mathbf{n}\in\mathcal{H}_N} e^{2f_\theta(\mathbf{n})}}, \quad p_\theta = |\Psi_\theta|^2$$

$$J_{mk} = \partial_{\theta_k} f_\theta(\mathbf{n}_m), \quad J \in \mathbb{R}^{M\times P}, \quad \mathbf{n}_m \sim p_\theta$$

$$\bar{J}_k = M^{-1}\sum_m J_{mk}, \quad (J_c)_{mk} = J_{mk} - \bar{J}_k$$

$$\bar{E} = M^{-1}\sum_m E_{\mathrm{loc},m}, \quad (E_c)_m = E_{\mathrm{loc},m} - \bar{E}$$

$$A = J_c/\sqrt{M}, \qquad e = E_c/\sqrt{M}$$

M counts retained configurations, not independent samples. P counts trainable parameters. These formulas use the real, positive-amplitude ansatz and fixed-N sector.

Imaginary time and the metric

$$\Delta \log \Psi(\mathbf{n}) \simeq -\eta \left[E_{\mathrm{loc}}(\mathbf{n}) - E \right]$$

$$d_{\mathrm{FS}}^2 \simeq \| A\, \Delta\theta + \eta e \|^2$$

$$S = A^{\mathsf{T}} A = J_c^{\mathsf{T}} J_c / M, \qquad f = A^{\mathsf{T}} e = J_c^{\mathsf{T}} E_c / M$$

$$\nabla_\theta E \simeq 2f, \qquad ds^2 = \Delta\theta^{\mathsf{T}} S\, \Delta\theta$$

SR matches a linearized variational step to imaginary-time evolution.

First-order wavefunction and second-order distance expansion; Monte Carlo estimates replace expectations. S is the real quantum geometric metric, not the energy Hessian.

Ridge SR as least squares

$$\mathcal{L}_\tau(\Delta\theta) = \|A\Delta\theta + \eta e\|^2 + \tau\|\Delta\theta\|^2, \qquad \tau > 0$$

$$A^T(A\Delta\theta + \eta e) + \tau\Delta\theta = 0$$

$$(S + \tau I_P)\Delta\theta = -\eta f$$

$$d = \mathrm{solve}(S + \tau I_P, f), \qquad \theta \leftarrow \theta - \eta d$$

The code assigns d as the gradient to native SGD.

The Euclidean parameter penalty stabilizes ill-conditioned directions and changes the step. No Adam is applied after SR; the learning-rate convention absorbs the energy-gradient factor of two.

The exact sample-space solve

$$T = AA^T \in \mathbb{R}^{M \times M}$$

$$(A^T A + \tau I_P) A^T = A^T (AA^T + \tau I_M)$$

$$d = A^T \mathrm{solve}(T + \tau I_M, e)$$

$$d = J_c^T \mathrm{solve}(J_c J_c^T + M\tau I_M, E_c)$$

Same full batch and ridge; use the smaller dense system.

For positive tau, this is the same regularized direction in exact arithmetic. Centered sample rank is at most min(P,M−1). No additional sample, parameter or rank truncation is introduced.

What ridge does to each mode

$$T = U\,\mathrm{diag}(\lambda_i)\,U^T, \qquad \lambda_i \geq 0$$

$$d = A^T U\,\mathrm{diag}[(\lambda_i + \tau)^{-1}]\,U^T e$$

$$A\Delta\theta = -\eta U\,\mathrm{diag}[\lambda_i/(\lambda_i + \tau)]\,U^T e$$

$$\lambda_i \gg \tau:\ \lambda_i/(\lambda_i + \tau) \simeq 1; \quad \lambda_i \ll \tau:\ \lambda_i/(\lambda_i + \tau) \simeq \lambda_i/\tau$$

Small-curvature directions are smoothly damped, not estimated reliably by assumption.

The ridge trades step fidelity for stability. A small sampled linearized step does not guarantee a small nonlinear wavefunction change or successful optimization.

Relation to published MinSR

$$d_{\mathrm{MinSR}} = A^{+} e = A^{T} T^{+} e$$

$$c = r_{\mathrm{pinv}} \max_i |\lambda_i| + a_{\mathrm{pinv}}$$

$$g_{\mathrm{soft}}(\lambda) = \{\lambda[1 + (c/|\lambda|)^6]\}^{-1} \quad (\lambda > 0)$$

$$d_{\mathrm{soft}} = A^{T} U \,\mathrm{diag}[g_{\mathrm{soft}}(\lambda_i)]\, U^{T} e$$

Our ridge uses 1/(λ+τ); the optional comparison uses the published soft filter.

Chen & Heyl, Nature Physics 20, 1476–1481 (2024), doi:10.1038/s41567-024-02566-1, Methods Eqs.20–23. Zero modes require explicit handling; actual SGD update is −ηd.

Cost, storage and precision

Dense formulation	Matrix to solve	Matrix algebra cost
Parameter space	$P \times P$	$O(MP^2 + P^3)$
Sample space	$M \times M$	$O(PM^2 + M^3)$
Full Jacobian retained	$M \times P$	$O(MP)$ storage
Numerical precision	Gram + solve: float64	Model derivatives may originate in float32

Costs exclude sampling and neural differentiation. Linear in P only at fixed M in the sample-space branch. Same-batch CPU audit: P=3681, M=256, primal/dual relative direction error 1.5×10^{-11}; not a long-run convergence proof.

Comparison with the 1D routine

1D: $(J^T J + \lambda_{\text{sum}} I) d = J^T E_c$

2D: $(J_c^T J_c / M + \tau I) d = J_c^T E_c / M$

$J^T J = J_c^T J_c + M \bar{J}^T \bar{J}$

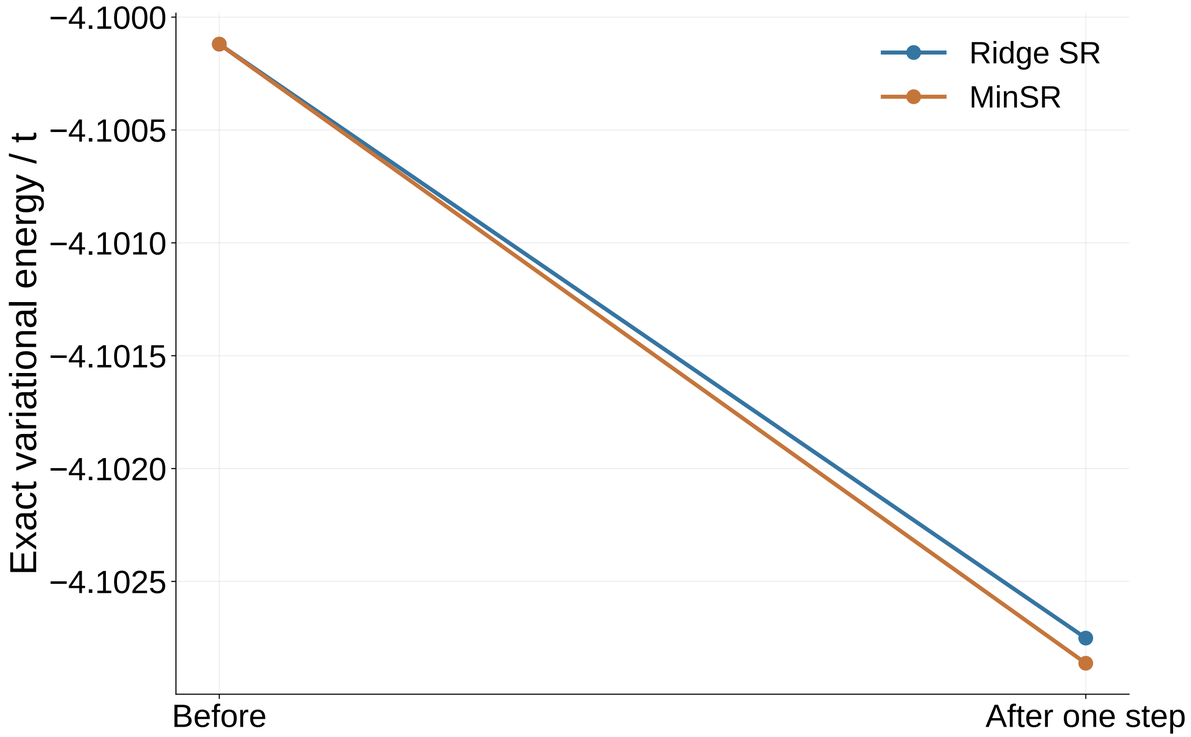
$\tau = \lambda_{\text{sum}} / M$ matches ridge units; centering still differs.

1D CPU code solves in float64; 2D solves on the training device in float64.

Changes to centering, ridge units and device are separate from Jacobian vectorization. The user accepted the 2D formulation. Gauge shifts can affect parameter differences without changing normalized-state tangents.

Small-system SR comparison

3×3 cylinder, RBM64; M=128, P=2989; one SGD step, η=0.001



Method	E after / t	ΔE / t
Ridge SR	-4.102751	-0.002631
MinSR	-4.102863	-0.002743

Ridge: τ = 0.001
MinSR: relative cutoff 10^{-12}, absolute cutoff 0

Same weights and 128 IID samples from current NQS probabilities. Energies evaluated over all 19,855 configurations (ED ground E/t = −4.385697). One batch and one step illustrate the methods, not convergence or superiority.

Simplest Transformer

Tokens and positions

```
occupation_embedding(n)
+ learned_x_embedding(x)
+ learned_y_embedding(y)

One token per physical site
Global attention over sites
```

Learned positions carry rectangular geometry.

One encoder block

```
TransformerEncoderLayer(
    d_model=32, nhead=1,
    dim_feedforward=64,
    dropout=0,
    activation="relu")

Mean pool → linear scalar
```

Width32 matches the periodic primary fits.

No phase/sign output is needed here. Neither mean pooling nor learned positions guarantee lattice translation symmetry.

Compare: 4×4 cylinder

Model	Parameters	Energy/site ± SEM	Gap	Neural gate
FC seed 654	9409	-0.474835 ± 0.000914	0.97%	pass
Jastrow seed 321	280	-0.459910 ± 0.001377	4.08%	baseline
Conv2D seed 654	17921	-0.476041 ± 0.000546	0.72%	pass
RBM seed 654	10448	-0.472243 ± 0.000770	1.51%	below target
Transformer seed 654	34433	-0.475023 ± 0.000734	0.93%	pass

Progress snapshot: confirmation seed 654 when available, screening seed 321 otherwise. A neural pass requires both seeds and all four targets. Jastrow is a fixed baseline, exempt from that gate.

Compare: 6×4 cylinder

Model	Parameters	Energy/site ± SEM	Gap	Neural gate
FC seed 654	48641	-0.497756 ± 0.000923	2.22%	below target
Jastrow retry seed 654	430	-0.476818 ± 0.001323	6.33%	baseline
Conv2D seed 654	26113	-0.506217 ± 0.000517	0.56%	pass
RBM128 retry seed 654	15608	-0.386679 ± 0.002684	24.04%	below target
Transformer seed 654	34561	-0.504490 ± 0.000396	0.90%	pass

6×4 CYLINDER: open x, periodic y; N=24, cutoff4, U/t=16.74. All rows use frozen seed654; gap is relative energy/site error to the same chi128 DMRG reference. Neural pass requires BOTH seeds: whole E/site and central-strip density/Gx/Gy point AND precision targets. Jastrow is the original displacement baseline, exempt from the neural gate. Budgets differ; no architecture ranking. J/RBM first attempts became nonfinite; selected fresh retries use smaller steps and fixed larger ridge (protocols retained).

Periodic VMC: local observables

2×3 torus	Mean D	Gx	Gy
ED	0.063387	0.298960	0.330886
FC exact	0.063092	0.298418	0.329289
FC MC321	0.06257 ± 0.00160	0.29142 ± 0.00588	0.33094 ± 0.00506
FC MC654	0.06204 ± 0.00207	0.30016 ± 0.00639	0.32442 ± 0.00668
Conv2D exact	0.063154	0.297654	0.330458
Conv2D MC321	0.06226 ± 0.00155	0.29523 ± 0.00609	0.32785 ± 0.00485
Conv2D MC654	0.06050 ± 0.00180	0.29882 ± 0.00585	0.31879 ± 0.00563

BOTH frozen seeds321/654 retained; exact refers to full426-state expectation of that NQS, while ED is the reference ground state. D=<n(n−1)>; sample composites precede blocking. Errors are conditionalSEM, not confidence intervals or mixing proofs.

Convolution + Transformer in 2D

Local features

One-hot grid

↓

3×3 Conv2d

↓

SiLU

32 channels. Periodic padding on both axes in these tests.

Global attention

One token per site

↓

4-head attention

↓

Feed-forward + norms

One encoder block; width32, FF64, ReLU, dropout0. No positional embeddings.

Scalar head

Flatten site tokens

↓

Dense32 + SiLU

↓

Scalar log amplitude

Physical order i=yLx+x. Dense head can break translation symmetry.

2D generalization of the existing 1D ConvTransformer: Conv1d becomes a boundary-aware Conv2d, followed by the same attention/flatten/dense pattern. Dropout is set to0 for deterministic VMC. Local convolution and global attention provide complementary features; improved accuracy is an empirical question, not guaranteed by this structure.

Hybrid: supervised ED benchmarks

System	Parameters	Fabs	δE/site / t	ED gate
2×3 torus	16225	0.9999999955	3.513e-08	pass
3×3 torus	19297	0.9995768953	0.001521	pass

Width32; one four-head block. Adam + normalized log-MSE,10000 updates, seed123; saved by log-MSE. Raw-ED checks in CPUfloat64. Both axes periodic, cutoff4. Tiny gate: Fabs≥0.9999 and δE/site≤0.001t; 3×3 working gate: Fabs≥0.99. A single fixed-budget fit tests this setup, not universal capacity.

Hybrid: fresh small-system VMC

2×3 torus	Exact E/t	Exact Fabs	Var(H)/t²
FC	-2.5734966	0.9999193	0.3786
Conv2D	-2.5798062	0.9999843	0.0603
Conv + Transformer	-2.5808163	0.9999979	0.01981

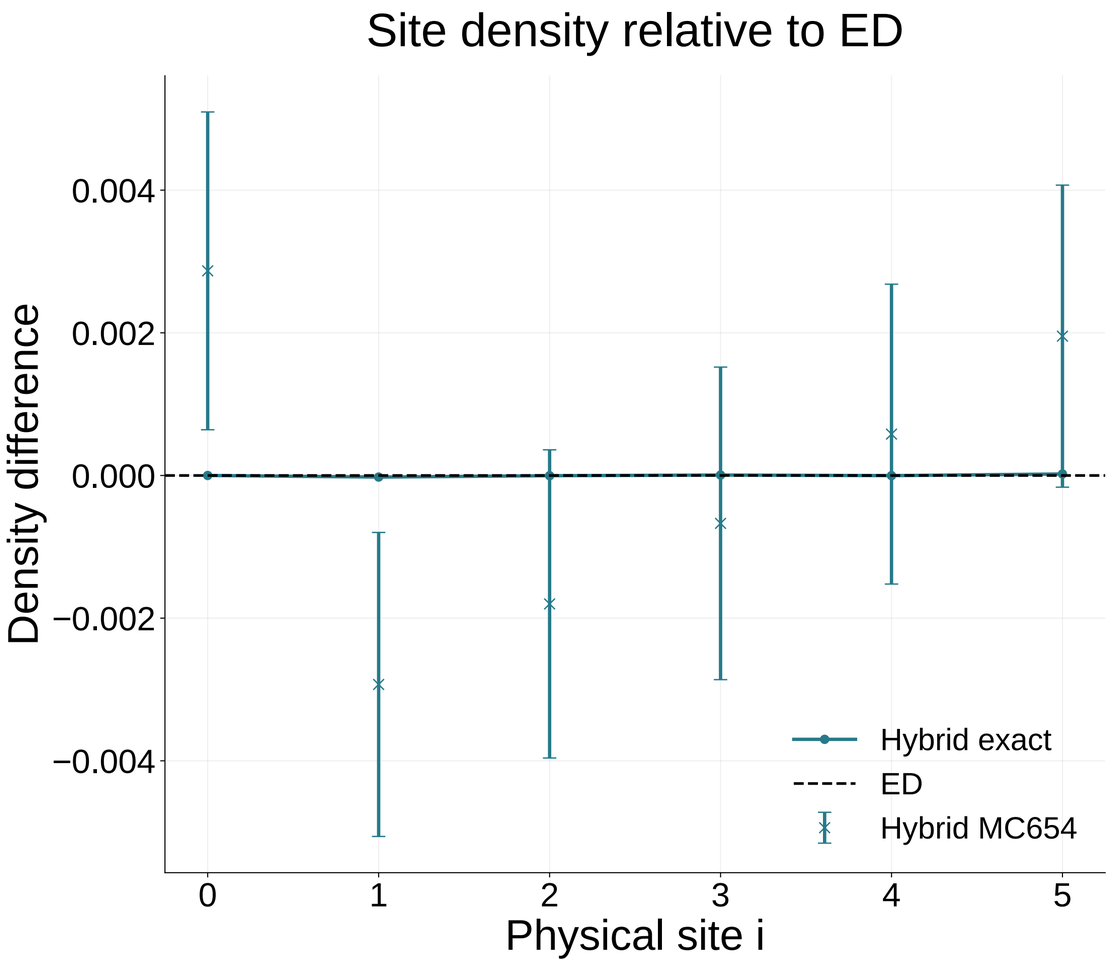
ED E/t=-2.581089512. Fresh seed123;500 stochastic ridge-SR + nativeSGD updates, M=1024; no ED labels or warm start. Exact full426-state final expectations. Hybrid runs GPUfloat32/TF32off with float64 SR; earlier FC/Conv run CPUfloat64. Different parameter counts and arithmetic: this is a functionality check, not a controlled ranking.

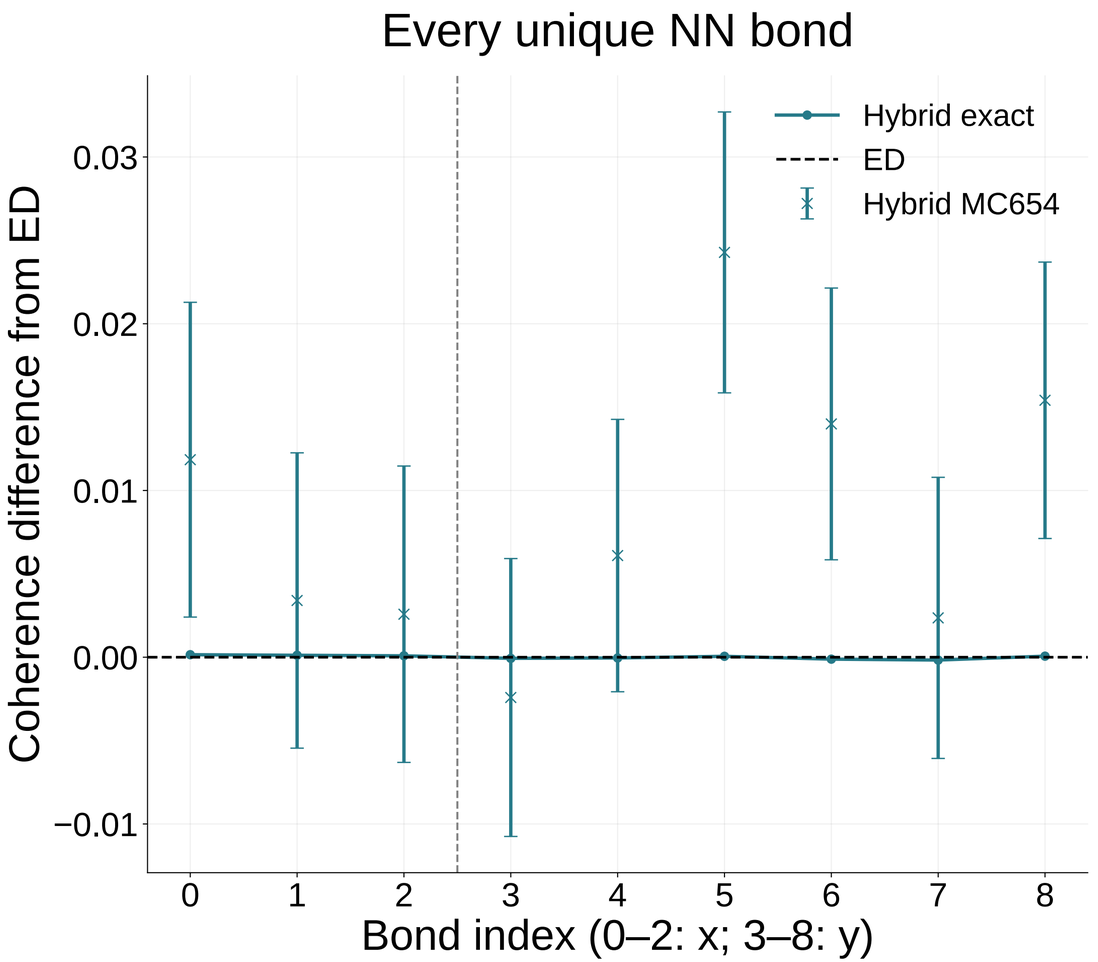
Hybrid: frozen observable checks

2×3 torus	E/t	Mean D	Gx	Gy
ED	-2.5810895	0.063387	0.298960	0.330886
Hybrid exact	-2.5808163	0.063396	0.299078	0.330839
Hybrid MC321	-2.58098 ± 0.00045	0.06263 ± 0.00159	0.29222 ± 0.00580	0.33109 ± 0.00495
Hybrid MC654	-2.58048 ± 0.00037	0.06649 ± 0.00169	0.30490 ± 0.00630	0.34084 ± 0.00555

BOTH frozen evaluation seeds retained, CPUfloat64,64walkers ×512retained steps, block64. D=<n(n−1)>; Gx/Gy average the three/six unique NN bonds. Composite each sample BEFORE blocking to retain covariance. Errors are max(blockSEM,between-chainSEM), conditional on mixing; exact-state error and MC scatter are separate.

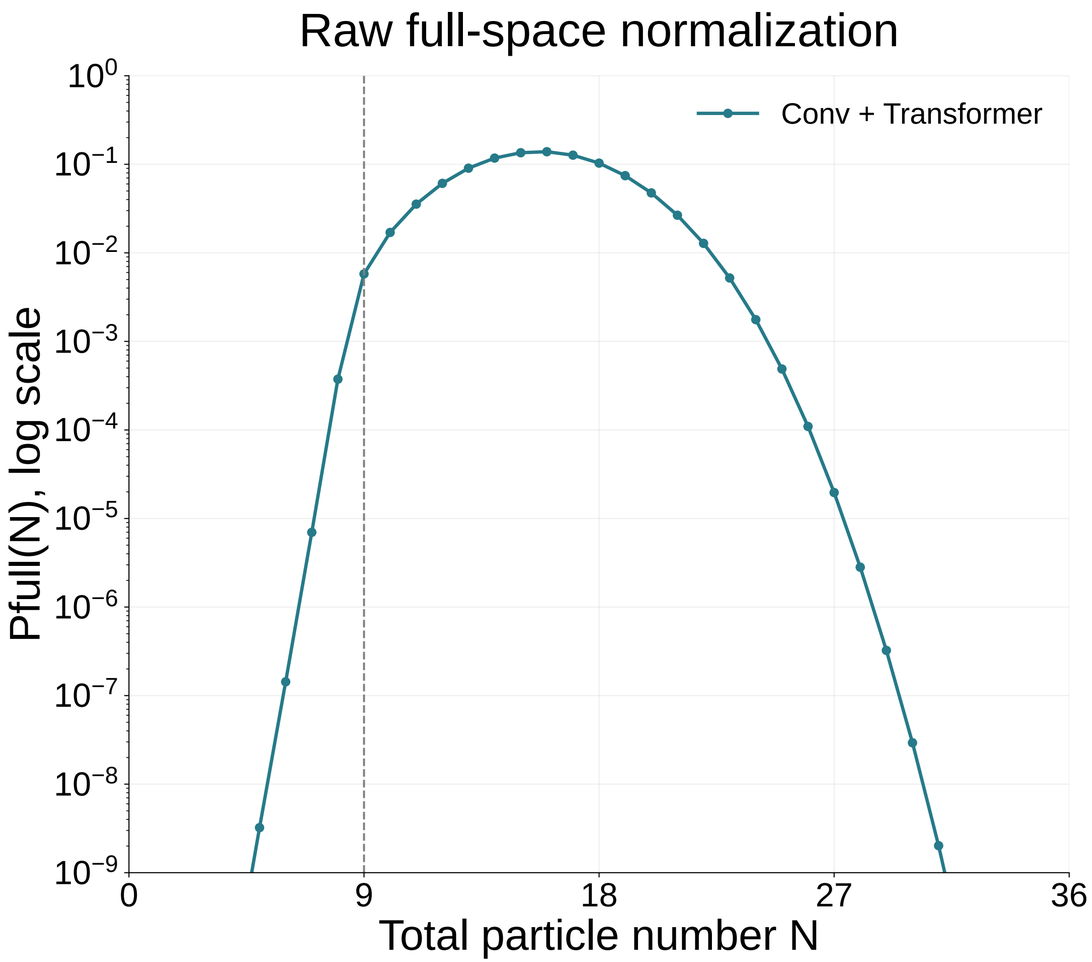
Hybrid: density and NN coherence

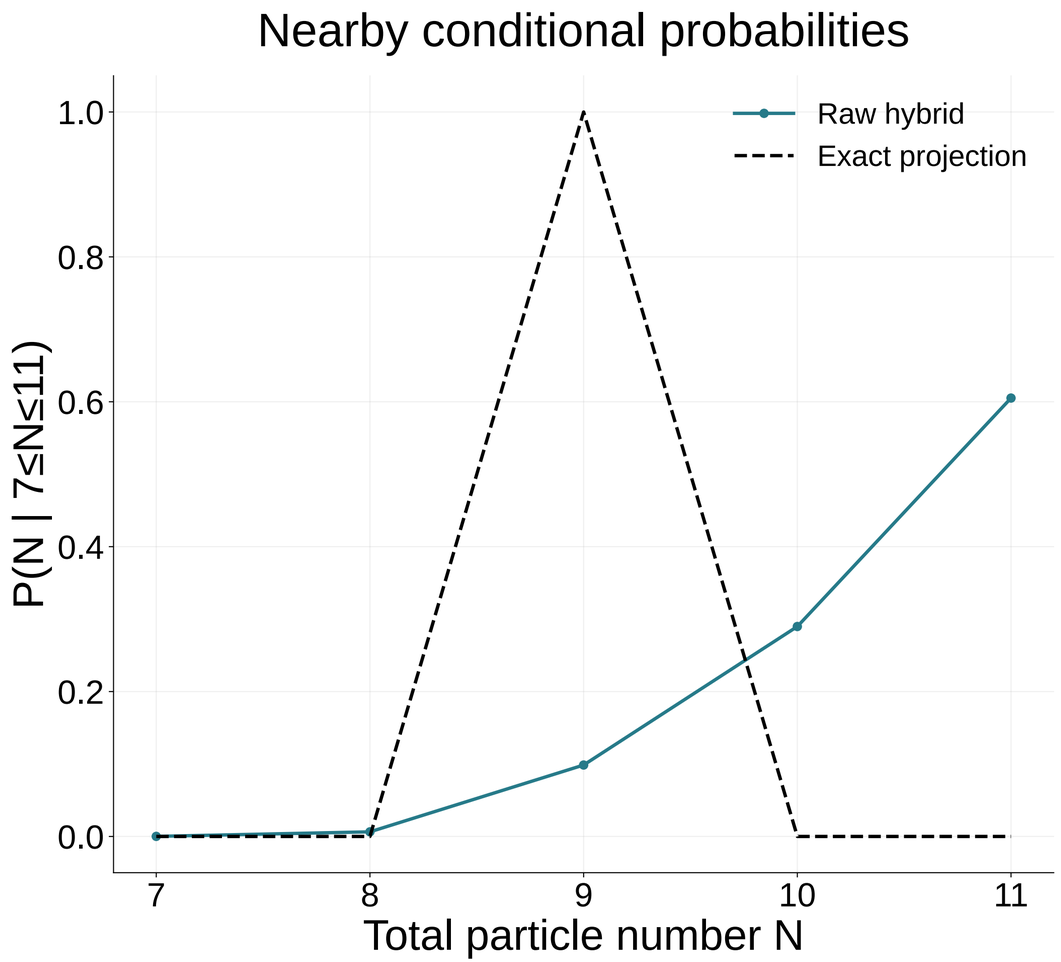




2×3 fully periodic, fixed N=6. Curves are exact expectations of the saved VMC state minus ED; crosses are frozenMC654 ± conditionalSEM. Includes periodic wraps and Hermitian bond estimators. Pointwise density, rather than its conserved mean, tests the learned profile.

Hybrid: raw neighboring-sector weights





Same supervised3×3 torus checkpoint, trained ONLY at N=9: F9=0.999577, raw Pfull(9)=0.0057816. All5^9 cutoff configurations enumerated; left-panel tails below10^-9 are clipped. Right panel renormalizes ONLY over N=7,…,11. Raw leakage is unconstrained extension, not canonical VMC contamination; exact number projection enforces physical support without retraining.